

HUMANITIES HUB WEB APPLICATION

Final Report

A Project

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By

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APPROVAL PAGE

PROJECT: HUMANITIES HUB WEB APPLICATION

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ABSTRACT

The Humanities Hub Web Application is a platform designed to utilize CRM technology to create a place to search for, learn about, and support humanities based businesses/organizations. This application utilizes search by filters or name to easily find a given institution, interactive maps to visualize a given location, and a rating system to share your opinion on an institution with the wider user base. The world is increasingly getting more disconnected and focused on technology, with many of the institutions included so far being ones that are not often talked about. This project aims to encourage interaction with these humanities based institutions and provide an easy-to-use web app to discover as well as support and even register institutions.

I. INTRODUCTION

The goal of this senior design project is to create a CRM-powered, full stack web application which provides a way to find, interact with, and register humanities based institutions. CRM refers to Customer Relationship Management, a style of technology that has been increasingly common among non-profit and corporate organizations as a sort of offshoot of more traditional forms of management of a given company. Salesforce, the CRM used in this project, provides the tools that will be spoken of going forward to build the web application. Full stack refers to the style of development in which both the front-end and back-end of an application are developed in tandem.

II. PRIOR RELATED WORKS

This project took inspiration from a variety of directory style websites made to display information on various sites and allow for individual interaction and review of them. One such example would be Yelp for businesses or Zillow for homes. While based around separate types of subjects, these websites share a purpose in allowing users to upload, view, interact with, and share specific locations like homes for purchase or businesses to visit.

III. TARGET AUDIENCE

Because of their universal use, there is no one specific age or gender demographic that websites like zillow or yelp targets, or that Humanities Hub would target. Rather, each one has a specific sort of audience that would interact with it. The Humanities Hub project would have a target audience of students as well as those more interested in the arts and humanities, as the project best suits itself to sharing information about low-cost or free institutions with some level of educational impact on those visiting.

IV. BACKGROUND STATISTICS

While this project was decided upon largely due to my own personal interest, there are some statistics to consider that offer contribution to the reasoning behind my project. The primary thing to consider is the fact that people in the United States, especially younger people, are spending less time outside and spending money than ever. Studies have been done in the US which found that “less than 20% of people in the country get outside more than once a week; 64% of people access the outdoors within 10 miles of their home; [12].” This is compounded with information in that same article, that speaks on how detrimental it is to one’s health to be away from green spaces or the outdoors. Beyond that, the Humanities Hub web app is focused for free or low cost humanities based institutions. This idea is supported by the current condition of unemployment, which is slowly climbing over 5% according to the federal reserve [13]. The rate of unemployment as well as the smaller share of wealth that younger generations have [14] mean that people overall have less wealth to be going out. Considering these statistics, it’s important to have resources like the Humanities Hub to find and support institutions which focus on low cost attractions or environments that can be both mentally stimulating as well as potentially educational.

V. DESIGN IDEAS

The design of this project was ultimately meant to prioritize ease of use and ensure that the target audience gets the necessary design and features to give a good user experience. The back-end aspects are also very crucial to the overall design, which will be more specifically discussed later in the report.

The first design choice I made was that of the foundational platform, my project is powered by Salesforce, meaning that I build my web app on the Salesforce CRM platform using

a mix of Apex Classes and Triggers to power the Visualforce page [3]. This means that the database issues were handled through the ‘Leads’ system and as custom objects which allowed more focus to be placed on how these objects were actually interacting with one another. The aforementioned ‘Apex’ based back-end code is a universal language used by Salesforce which functions much like java with features similar to html, css, and javascript [3]. This style of CRM technology seems of much use in non-profit organizations as well as educational institutions to varying degrees, which plays into the idea that the Humanities Hub project would be more charitable than focused on profits.

The web page itself is a directory with functionality reminiscent of the aforementioned websites of inspiration, using a top bar that holds all the options while the directory is a set of horizontal, selectable institution cards which can be expanded with details or interacted with directly. The top bar, much like many other similar sites, includes a text search bar, filter by category, filter by city, and filter by state. There’s also a search button, though the enter key works just fine, a clear button to clear all search parameters, and the register button which redirects to the registration form. This all is a well established style of design for a directory style website like the Humanities Hub Project, making it easy to use for new users and intuitive in design.

Some further consideration was made on the platform, as it is currently only fully functional through a webpage, but could be converted to an android style layout with minimal issues. The Humanities Hub project, much like yelp or zillow mentioned before, is meant to be a searching site where people would go to gather initial information or do in depth research before deciding on destinations or choices. This sort of purpose and use means that a webpage

accessible via some sort of computer or phone would be the first priority for the project, and thus why it was developed through visualforce for a webpage design format.

VI. USE CASE DIAGRAM

The Humanities Hub project is a software project, and thus there are various diagrams we can use to help illustrate its functionality and design. One such diagram being the Use-Case Diagram, which illustrates how users will interact with the project software. The Use-Case diagram was created using an online drawing tool for visualization, ‘VP Online’ [2]. In this example, there are two actors listed, the first being a standard user, and the second being the Org representative.

The user will first enter the site, gaining access to the directory where they might search, filter, and generally browse the already established institutions. The user has several options here where they might interact with each individual institution through selectable button widgets to access the institution’s website, donation page, contact email, or to enter the details page of the individual institution.

When a user selects into an institution, the user then gets a variety of options available to them within the expanded page showcasing further details on the institution, staff, and location. There is an interactive map which will be spoken of in more detail later, as well as a rating system with persistent scoring tied directly to the individual institution, and the same button widgets as before to be redirected to the email, website, or donation page of a given institution.

The last major option of the user, when at the main directory, is to select the ‘Register Now’ button and become an Org representative user. The Org representative user has a simple use-case flow, with them being able to register new institutions by filling out a form, submitting the form, receiving a confirmation email, and then being sent back to the directory through a

‘thank you’ page. The org representative becomes a standard user once they re-enter the main directory.

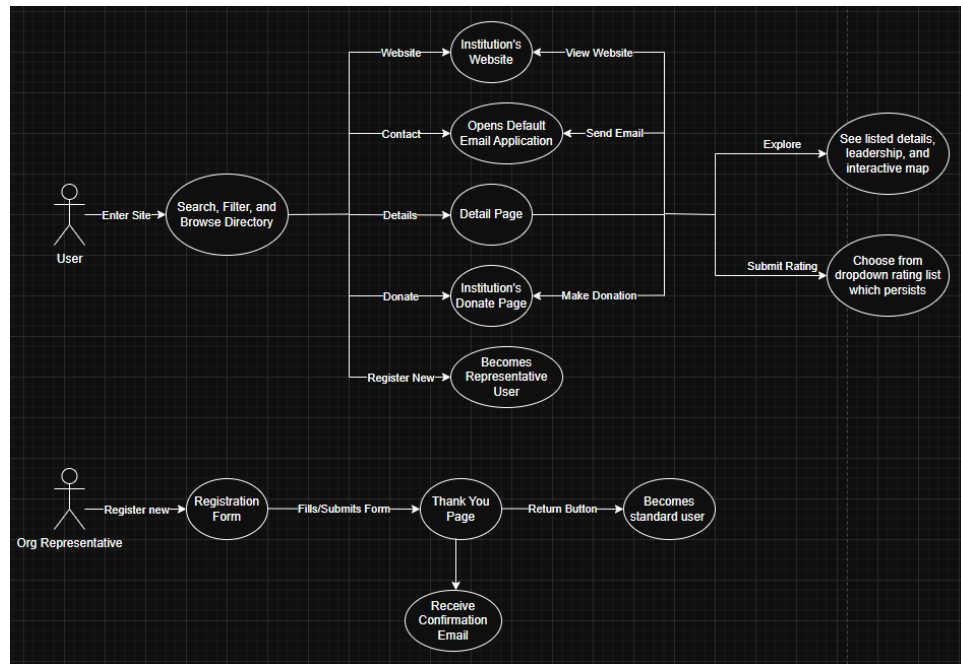


Fig. 1 - Use Case Diagram

VII. SEQUENCE DIAGRAMS

The below two sequence diagrams serve as visualizations of the process of registration for the Humanities Hub project, split between the user side and sys admin side. Both sequence diagrams were created through an online flowchart maker [1]. In general, the registration process is a mostly automated process in which Apex Triggers, automated scripts, do most of the heavy lifting for turning a registration form into a lead object and then eventually into a full pairing of institution parent and staff child objects.

This first sequence diagram, figure 2, focuses on the ‘Org User’ actor, representing the user who has gone to register an institution through the registration button selectable in the main directory. This registration form is hosted on Github Pages, allowing for the user to enter all the

necessary information to the form. This form then POSTs the information directly to salesforce through the Web-To-Lead service, which is automated through the Apex Trigger 'LeadAutomation' file which creates the lead object in salesforce for review. The Github Pages registration form redirects the user through a hyperlinked 'Thank You' page back to the directory. The org user receives a confirmation email and then is once again a standard user in the web app to further explore the directory until their registered institution is confirmed and implemented into the directory.

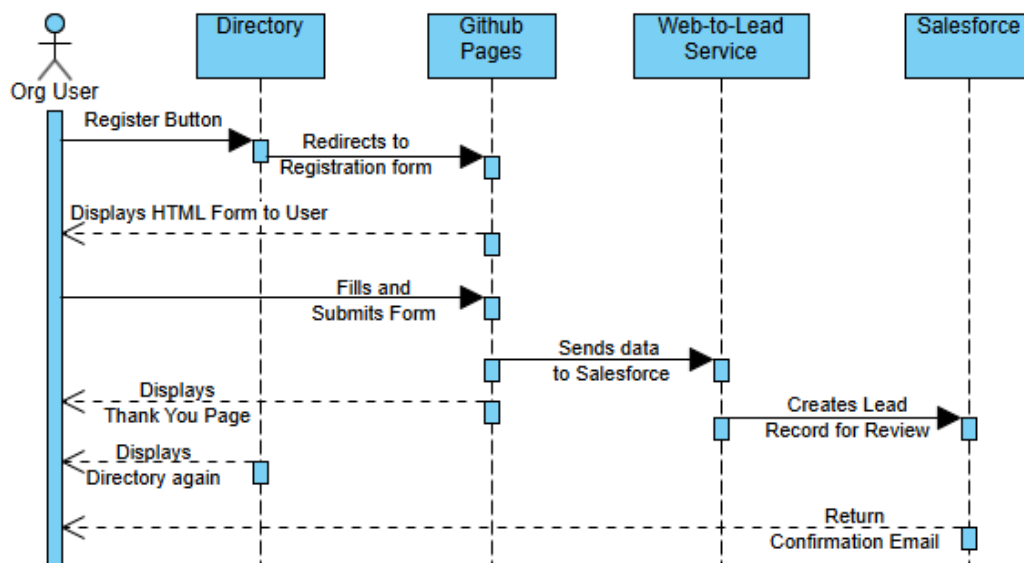


Fig. 2 - Sequence Diagram (Org Registering User)

The second sequence diagram below, figure 3, illustrates the system administrator's point of view of the registration process, beginning after the LeadAutomation trigger is used to turn a submitted registration form into a Lead object. The system admin first reviews and validates the Lead data, checking for misspellings or misinputs, then shifts the progress of the lead to trigger the Apex Geocode Trigger which accesses OpenStreetMap [15] through the Nominatim API to generate the coordinates which allow the interactive map to function per institution. The system

admin can then turn the Lead into an official institution object which is automatically uploaded with the rest of the institutions to the directory once its status is set to ‘complete’.

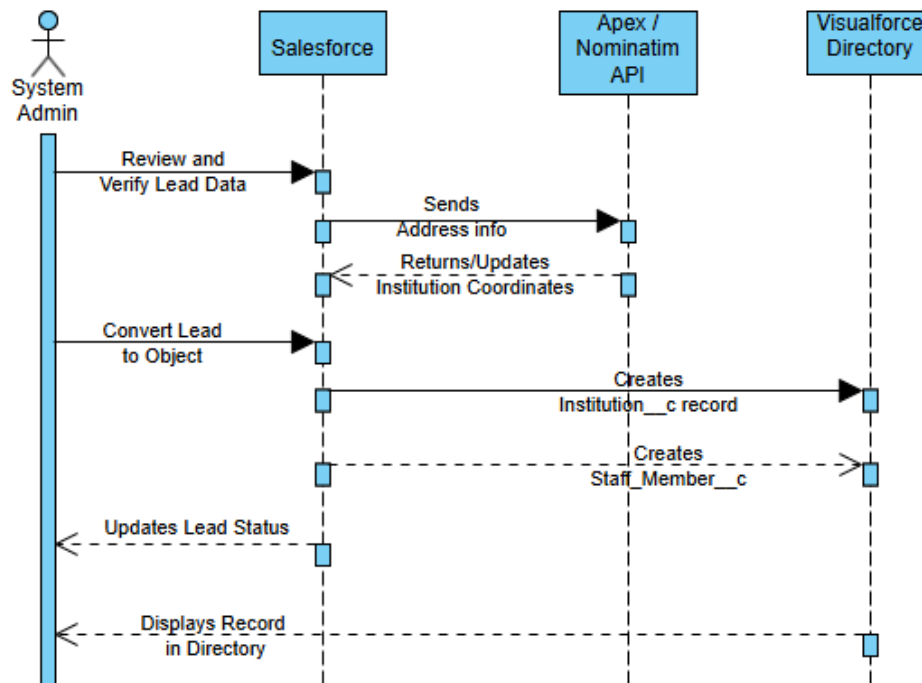


Fig. 3 - Sequence Diagram (Sys Admin)

VIII. CLASS DIAGRAM

The last diagram is that of a UML class diagram for the project as a whole, and allows a full visual representation of how the project’s classes, functions, and objects work together to create the web page. This is figure 4, which was created using an online drawing tool, ‘VP Online’ [2]. The general flow of the class diagram has the Institution__c custom object at its center as all other functions of the project are tied to the individual institutions.

The first major class and theoretical entry point into the system is the registration form hosted on github pages, which is what creates the Leads for the institution with the help of the LeadAutomation trigger. The lead creation triggers the thank you page via the registration form, and the Lead is eventually turned into the Institution custom object.

The institution custom object, as mentioned before, is the core of the software, and it has two child objects, the first being Staff_Member custom object, which is created alongside the institution as a part of the Lead to Institution process. The Institution custom object's second child object is that of the Institution_Rating custom object, which is what holds the specific rating of each institution through a COUNT, SUM, and FORMULA set of fields, all used to produce an average persistent rating to display via the Institution object.

An important piece of the Humanities Hub project is the flow for geocoding, an important component which ensures that simple addresses given via the registration form can be displayed on the details page of each Institution. When a lead is being turned into an Institution, as well as on every instance of the object being updated, the Apex Trigger for InstitutionGeocode is activated, which calls the InstitutionGeocodeHelper Apex Class to make use of OpenStreetMap API to turn the address into a set of longitude and latitude coordinates to fill the location field of each institution [15].

And finally, there is the visualforce page, which receives the many Institutions and their data for display. The visualforce page is driven by the largest class of this project, the Apex Class of InstitutionDirectoryController. As the name suggests, this class drives most all the functionality of the software and specifically the directory page. The visualforce page can be thought of as the primary front-end while the InstitutionDirectoryController can be mostly thought of as the primary back-end.

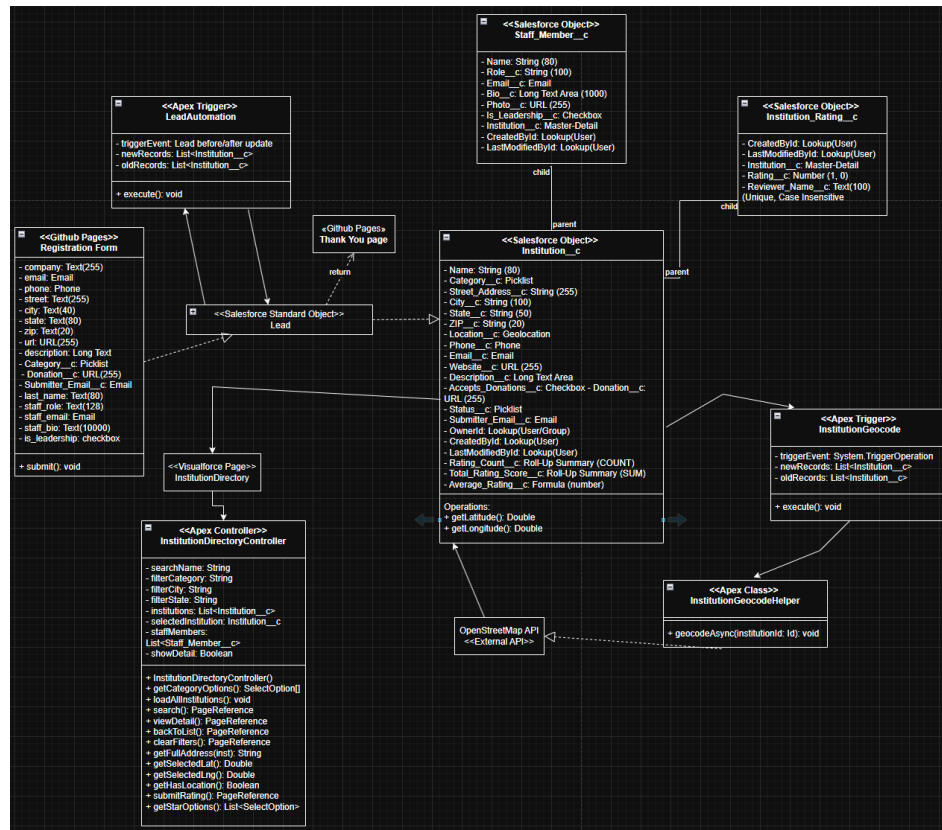


Fig. 4 - Class Diagram

IX. FRONT-END DEVELOPMENT

The front-end of the Humanities Hub project was created through Visualforce Apex code. This is a Salesforce specific language that takes inspiration from common and versatile languages like Java for classes and triggers, and html/css for visualforce page functionality, and even javascript-esque script calling for API communication. Both the main directory and details pages are created through Visualforce code which organizes everything through html style code and gives functionality to the JavaFX style widgets and functions. It's important to note that Apex code, and thus salesforce applications as a whole, are not designed for only one system, and can be accessed via both mobile and desktop so long as the webpage itself is still accessible.

Below is the main directory page, figure 6, created through visualforce, which allows for full functionality of that top bar where there's the search by name, category drop down, search by city, search by state, as well as the three buttons beside them. These are all aligned through code that reminded me of JavaFX as well as CSS, and I elected to utilize those tools to make a directory much like any other search directory style website. Each individual institution is displayed with buttons, and each one can be selected via the details button for more information.

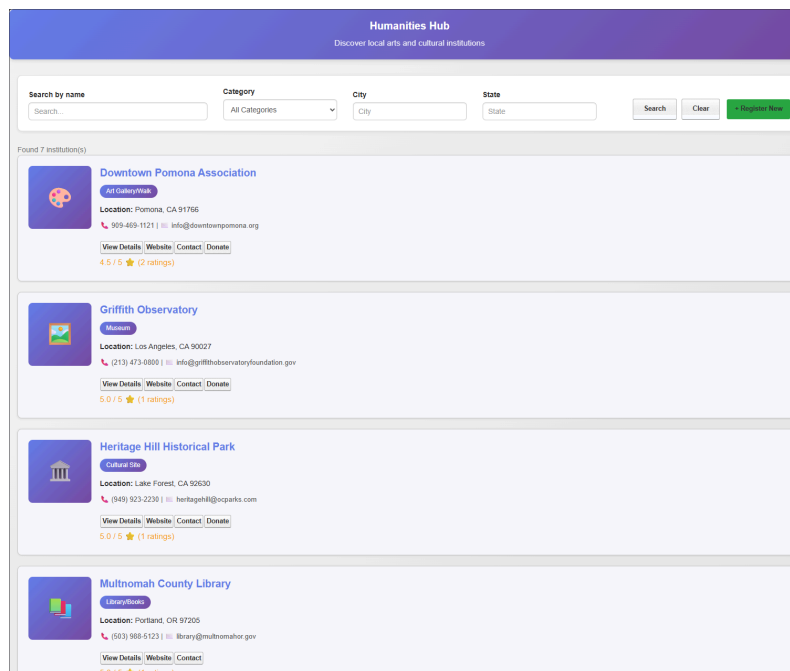


Fig. 6 - Main Directory Page

After selecting the details button on one of the institutions, the user is entered into a uniform webpage loaded with the information of the specific selected institution. Figure 7 shows one such institution's detail page as reference. The category of the institution is what determines the image shown at the top, though this placeholder image can be traded out for real images off of image hosting sites that my salesforce refused to properly communicate with, which will be discussed in more detail in the APIs section.

This example shows the Downtown Pomona Association example institution, which hosts monthly pomona artwalks among other events, though this is specifically promoting the artwalk part of the institution [5]. All information gathered through the registration form and then processed will be displayed here as the institution and staff_member objects, providing just enough details for the user to decide whether or not they want to delve further into the institution's website or know for sure if they have the right location. Apex visualforce code allows for these various details to be organized neatly, and uses OpenStreetMaps [15] as well as the rating system to give the user an interactive experience.

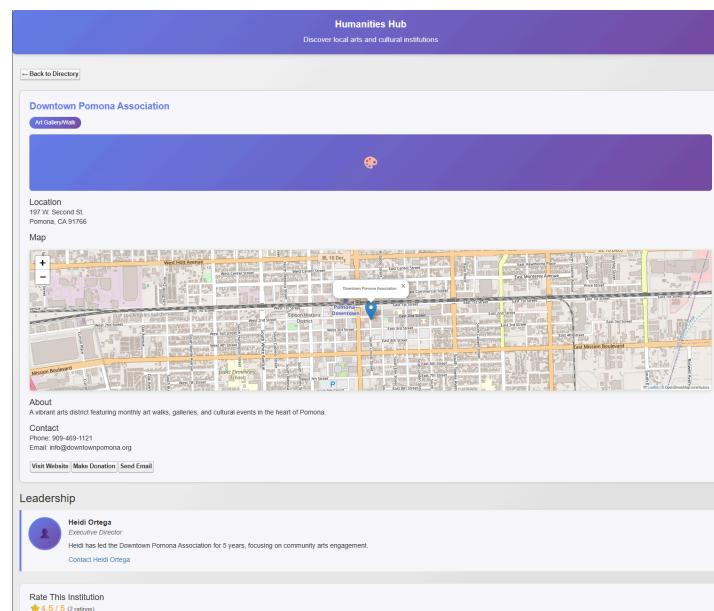


Fig. 7 - Details Page

X. GITHUB PAGES

As mentioned previously, the Humanities Hub project hosted a small number of webpages through Github Pages, which is a free webpage hosting service of Github that allows for webpages to be made and interact with other services [16]. Below, in figure 8, the 'Thank You' page is shown, which is the simpler of the two main webpages created for this project. This

page makes use of the same theme as the main website and has the simple duty of informing the user of the submission and redirecting them to the main directory to continue use of the main webpage.

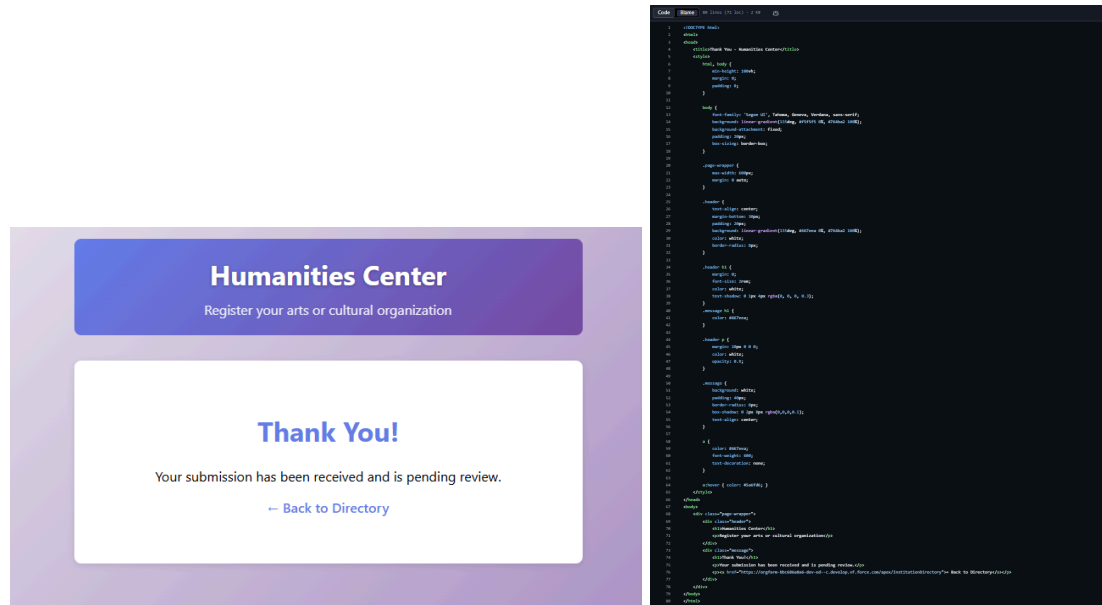


Fig. 8 - Thank You Page/Code

The other and more important webpage made through Github Pages is the actual registration form itself. This registration form, shown in figure 9, also follows the main theme of this project of the gradient purple coloration, which is just personal preference. The registration form is long and cohesive, providing an application-esque form to be filled out by an org representative with all the relevant information for their Institution to be included in the main directory.

The registration form is roughly split into two parts as it will ultimately create two custom objects, the parental Institution custom object and the child Staff_Member object. The first several questions ask only about the institution, including for its email, phone, and other contact or important information. This includes a variety of drop down menus as well as open writing menus which all show examples of text to help the user input their data in a way that will

make automation as smooth as possible. There is then a small space that splits the form, informing the user that the following questions are now for the ‘Primary Contact Person’, which in my examples I made as leadership individuals, but in practice this person could be anyone and there is no requirement to select the ‘In Leadership’ checkbox.

Upon pressing the ‘Register Organization’ button, the user is then redirected to the ‘Thank You’ page while all of their information POSTs to salesforce. To ‘POST’ is to push information directly from the registration form on github to the Humanities Hub organization’s Web-To-Lead endpoint. This Web-to-Lead endpoint is how the salesforce organization communicates with the registration form, turning the data into a raw Lead which holds all the data. As mentioned somewhat before, this Lead is then processed via the LeadAutomation Apex trigger to create both the Institution custom object as well as the childed Staff_Member custom object.

Humanities Center

Register your arts or cultural organization

Organization Name *
e.g.: Dunder Mifflin Inc.

Phone
e.g.: #####-#### or ###-###-#### or ###.###.####

Institution Contact Email *
e.g.: name@gmail.com

Organization Description
Tell us about your organization's mission and activities...

☒ This organization accepts donations

Donation Page URL
https://www.example.com/donate

Street Address *
197 W. Second St.
Apt 4

City *
Pomona

State *
Select State

ZIP Code *
91766

Website
https://www.example.com

Category *
--Select Category--

Donation Page URL
https://www.example.com/donate

Street Address *
197 W. Second St.
Apt 4

City *
Pomona

State *
Select State

ZIP Code *
91766

Website
https://www.example.com

Category *
--Select Category--

Primary Contact Person
This person will be added as the first staff member and primary contact for your organization.

Full Name *
e.g.: Dr. Jane Smith, Director of Marketing

Your Role/Title *
e.g.: Executive Director, Curator, Volunteer Coordinator

Your Email *
you@example.com

Brief Bio (optional)
Tell us about your background and connection to this organization...

☒ This person is in a leadership position

Register Organization

Fig. 9 - Registration Page

XI. APEX CLASSES

Apex Classes are one of the biggest parts of the Humanities Hub project's construction, they are reusable blocks of code wrapped into classes that define what the app can actually do, its functions. This includes the main directory driver class as well as helper classes for the triggers. Classes function much like toolboxes for the app to utilize. This project focuses on two custom Apex classes which will be described below.

The first Apex class is that of the InstitutionGeocodeHelper, this class acts as the helper to the InstitutionGeocode trigger, which will be further defined in the next section. The InstitutionGeocodeHelper class focuses on gathering the address information within the Lead created from the registration form and calling on OpenStreetMaps API to convert the address into a set of Longitude and Latitude coordinates for the function of the interactive map within the details page of each institution.

The other Apex class, and the one that primarily drives the visualforce web application's functionality, is that of the InstitutionDirectoryController. This holds most of the logic for the app and is what makes everything work. This not only includes the functionality of widgets or interactive areas, but also core functionality like the rating system displayed in figure 10. As seen in figure 10, the code for the apex driver class is similar to Java, with this method being one to gain access to star options rather than an open input area for the rating system. This static option style of rating was chosen due to its reliability with the formulas used in the Institution_Rating child object attached to each individual institution. The total rating of an institution is calculated through this near invisible object with only the average rating being displayed directly. The average rating is gathered by using COUNT to gather the number of ratings, then SUM to add

them all up, and finally FORMULA using a standard averaging formula to display them for each institution.

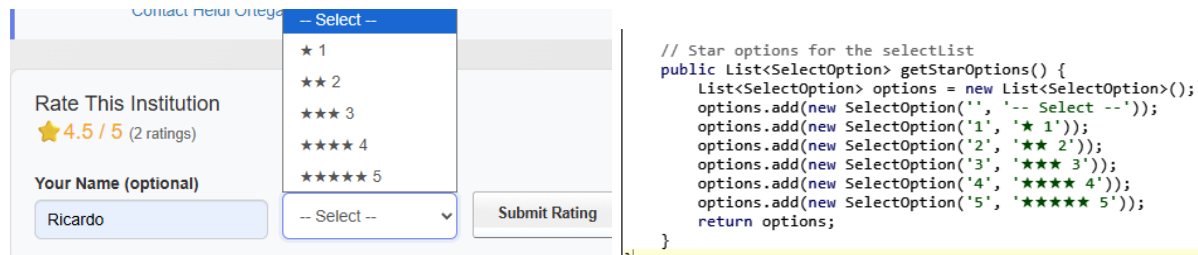


Fig. 10 - Rating System

As mentioned above, the InstitutionDirectoryController also manages all other functionality of the web app, that includes the filtering system shown in figure 11, a core component of this project. The searches and filters were established using code reminiscent of SQL as fields of objects are stored within salesforce much like a standard database, allowing for precise queries to be applied through either drop down menus or direct written requests to be entered into queries for the database. As seen in figure 11, the city is filtered with the text ‘Pomona’, the state is filtered with the text ‘CA’, and the dropdown menu is selected onto the ‘Art Gallery/Walk’ category. And even without the name being entered, only the Downtown Pomona Association is shown [5]. It’s important to note that there are a variety of pomona locations included in my examples, but only one within this specific category. Others include the city of Pomona library [11] and the Pomona Hope foundation [10].

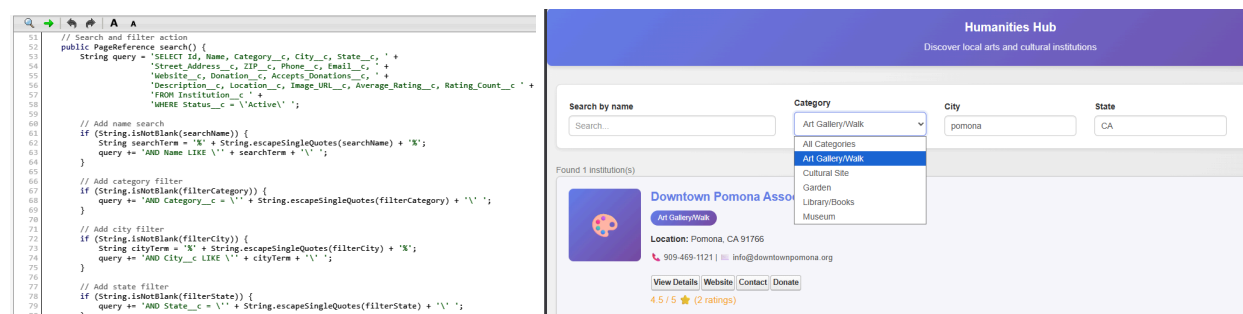


Fig. 11 - Filter System

XII. APEX TRIGGERS

Whereas Apex Classes are the ‘doers’, the actionable portions of code that can be utilized to give functionality to the application, Apex Triggers are more so the ‘watchers’, which watch for and then react to the Humanities Hub project’s data and events. There are two main triggers created for this project, both of which have been mentioned in this report.

The first trigger to speak of is the InstitutionGeocode Apex Trigger, this trigger watches for the events of an institution being created or edited, ensuring that when those occur then the address will be processed through OpenStreetMaps API and converted into accurate longitude and Latitude coordinates for the functionality of the interactive map.

The second and arguably more important trigger is the LeadAutomation trigger which focuses on handling the process of turning the Lead into an Institution object and its child Staff_Member object. Figure 5 below showcases what the raw Lead details page looks like, having a host of unused fields and acting only as a stop on the way from registration form to Institution and Staff_Member Object. The job of turning a Lead into the aforementioned custom objects revolves around matching the data that the Lead captured and inputting it into the appropriate fields within the newly made custom objects. This process is only partially automated, as the System Admin still needs to manually adjust and ensure correctness due to API troubles which will be discussed in the next section.

Lead
Delta Pick Mello

+ Follow New Case New Note Submit for Approval

Lead Owner	Phone
Ricardo Gaeta	(916) 868-7059
Name	Mobile
Delta Pick Mello	
Company	Fax
Sacramento History Museum	
Title	Email
	rgaeta209@gmail.com
Lead Source	Website
	https://sachistorymuseum.org/
Industry	Lead Status
	Open - Not Contacted
Annual Revenue	Rating
Category	No. of Employees
Museum	
Submitter Email	
rgaeta209@gmail.com	
Is Leadership	
✓	
Donation	
https://sachistorymuseum.org/get-involved/donate/	
Address	
1011 Street	
916 Sacramento, California 95814	
United States	
Product Interest	Current Generator(s)
SIC Code	Primary
Number of Locations	
Created By	Last Modified By
Ricardo Gaeta, 4/30/2026, 10:47 AM	Ricardo Gaeta, 4/30/2026, 10:50 AM
Description	
The Sacramento History Museum is dedicated to telling the story of Sacramento's people and place. From the home to its first people, to the cultures of its current population, you will discover the impacts of California's Gold Rush – the massive migration of people from all over the world to the Sacramento region in search of opportunity, prosperity, and inspiration. The result is a region with richly diverse communities and cultures; an agriculture and transportation hub; and capital of the State of California, the 39th largest economy in the world.	
Staff Bio	
Proud administrative head of the Sacramento History Alliance (SHA) is a nonprofit, 501(c)(3) organization dedicated to making Sacramento's history accessible to all.	

We found no potential duplicates of this Lead.

Campaign History (0)

Fig. 5 - Leads Page

XIII. APIs

APIs have been a major hurdle throughout the process of the Humanities Hub project, a hurdle that was not fully overcome. Initially, the idea of the project was to be a far more fluid, reliable, and automated registration process through the work of python's FastAPI. FastAPI was meant to communicate in the place of the Web-To-Lead system of salesforce, instead taking the information from the registration form and directly creating the new Institution and Staff_Member objects for a given institution through that, and simply marking it's status as 'pending' to ensure it wasn't immediately pushed to the directory without review. FastAPI communicates with both Github through the registration form, as well as the trouble child of SoapAPI. SoapAPI acts as the receiver to scripts and API calls, and unfortunately, this receiver API was never able to function for me in any capacity. Whether through faults of skills or by misfortune in the free edition of the organization (app) of Humanities hub, or even because the organization is technically registered through the CSUDH student email, this caused a severe

limitation on more exotic or impressive functionality. This also served as a large waste of time, with my first progress report being nothing more than a prototype of the registration page and thank you page before I turned my focus to doing what I could directly in salesforce.

The only other APIs that were used to any significant degree, and worked hand in hand, is that of OpenStreetMaps API [15] which has been mentioned before, and [leaflet.js](#) API [16] which uses the coordinates gathered in the prior API to display the actual interactive map that has been mentioned before.

XIV. CONTRIBUTIONS

Creating the Humanities Hub web application was a testament to my, admittedly misplaced, dedication to accomplishing the goal of utilizing a professionally used technology, like the CRM that is Salesforce, to create something new and interesting that could have some level of real world implications. This was a process of research, technical skill, creativity, and expression of my deep interest in more humanistic organizations and locations. Overall, while I'm not fully satisfied with the result of my hard work, I am proud of myself for completing it to this degree and when considering recent circumstances in my life and community.

XV. CONCLUSION AND FUTURE WORK

In conclusion, the Humanities Hub project was made to allow for students, academics, hobbyists, or simply others who share the passion, to explore, search for, interact with, and even register Humanities Based Institutions. The application is meant to help provide humanistic alternatives to standard daily life, to help introduce and spread local organizations within these categories with others, and to share your thoughts via rating.

For the future prospects of this project, it would likely be best to take the ideas and features here and export them to a secondary base system to build on in order to allow for more complete and reliable API functionality. Assuming this, I would then plan to make the registration system seamless, as well as implement a full user profile system so as to track who is registering what institutions and rating what institutions, likely through standard database solutions that can interact with web applications like firebase. Another major feature that I would've liked to implement would be a 'Discover Local' feature, which would use the user's location to provide suggestions for nearby registered Institutions to explore.

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XVII. CODE

Visualforce (webpage) code:

```
<apex:page controller="InstitutionDirectoryController"

    sidebar="false"
    showHeader="true"
    standardStylesheets="true"
    docType="html-5.0">

    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"/>

    <apex:stylesheet value="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css"/>

    <style>

    body {
        background: linear-gradient(135deg, #f5f5f5 0%, #e0e0e0 100%);
        min-height: 100vh;
        margin: 0;
        padding: 0;
    }

    .hub-header {
        text-align: center;
        margin-bottom: 30px;
        padding: 20px;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white;
        border-radius: 8px;
        box-shadow: 0 2px 8px rgba(0,0,0,0.15);
    }

    .hub-header h1 {
        margin: 0;
        font-size: 2rem;
        font-weight: 600;
    }

    .hub-header p {
        margin: 10px 0 0 0;
        opacity: 0.9;
        font-size: 1.1em;
    }

    <!-- Search panel -->
    .search-panel {
        background: #ffffff;
        padding: 20px;
        border-radius: 8px;
        margin-bottom: 25px;
        box-shadow: 0 2px 6px rgba(0,0,0,0.08);
        border: 1px solid #ddd;
    }

    <!-- Institution cards -->
    .institution-card {
        display: flex;
        align-items: flex-start;
        gap: 16px;
        border: 1px solid #ccc;
        border-radius: 8px;
        padding: 20px;
        margin-bottom: 20px;
        background: #f8f9fa; /* Light gray instead of pure white */

        transition: all 0.3s ease;
        box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    }

    .institution-card:hover {
        background: #ffffff; /* White on hover */
        box-shadow: 0 8px 25px rgba(102, 126, 234, 0.25); /* Purple-tinted shadow */
        transform: translateY(-2px); /* Slight lift effect */
        border-color: #667eea;
    }

    .institution-name {
        font-size: 1.4em;
        font-weight: bold;
        color: #667eea; /* Match gradient purple */
        margin-bottom: 8px;
    }

    .category-badge {
        display: inline-block;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white; /* White text on purple */
        padding: 4px 12px;
        border-radius: 20px;
        font-size: 0.85em;
        font-weight: 500;
        margin-bottom: 12px;
    }

    .contact-info {
        color: #555;
        font-size: 0.95em;
        margin-top: 12px;
    }

    .action-buttons {
        margin-top: 18px;
    }

    .staff-card {
        border-left: 4px solid #667eea;
        padding-left: 15px;
        margin: 12px 0;
        background: #f8f9fa;
        padding: 15px;
        border-radius: 0 8px 8px 0;
        display: flex;
        align-items: flex-start;
        gap: 15px;
    }

    .no-results {
        text-align: center;
        padding: 50px;
        color: #666;
        background: white;
        border-radius: 8px;
        box-shadow: 0 2px 6px rgba(0,0,0,0.08);
    }

    #map {
        height: 300px;
        width: 100%;
        border-radius: 8px;
        border: 2px solid #667eea;
        margin: 15px 0;
        box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    }

    /* Form styling improvements */
```

```

form-control {
  border-radius: 6px;
  border: 1px solid #ccc;
  padding: 10px;
  transition: border-color 0.2s;
}

form-control:focus {
  border-color: #667eea;
  box-shadow: 0 0 3px rgba(102, 126, 234, 0.1);
}

.btn-primary {
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  border: none;
  border-radius: 6px;
  padding: 10px 20px;
  transition: all 0.2s;
}

.btn-primary:hover {
  transform: translateY(-1px);
  box-shadow: 0 4px 12px rgba(102, 126, 234, 0.4);
}
</style>

```

```
<!-- Header -->
```

```
<div class="hub-header">
```

```
<h1>Humanities Hub</h1>
```

```
<p>Discover local arts and cultural institutions</p>
```

```
</div>
```

```
<div class="container-fluid">
```

```
<apex:form >
```

```
<!-- LIST VIEW -->
```

```
<apex:outputPanel rendered="{!!showDetail}">
```

```
<!-- Search & Filter Panel -->
```

```
<div class="search-panel">
```

```
<div class="row" style="display: flex; align-items: flex-end;">
```

```
<div class="col-md-3">
```

```
<label>Search by name</label>
```

```
<apex:inputText value="{!searchName}"
```

```
styleClass="form-control"
```

```
html-placeholder="Search...">
```

```
</div>
```

```
<div class="col-md-2">
```

```
<label>Category</label>
```

```
<apex:selectList value="{!filterCategory}" size="1"
```

```
styleClass="form-control">
```

```
<apex:selectOptions value="{!categoryOptions}">
```

```
</apex:selectList>
```

```
</div>
```

```
<div class="col-md-2">
```

```
<label>City</label>
```

```
<apex:inputText value="{!filterCity}"
```

```
styleClass="form-control"
```

```
html-placeholder="City">
```

```
</div>
```

```
<div class="col-md-2">
```

```
<label>State</label>
```

```
<apex:inputText value="{!filterState}"
```

```
styleClass="form-control"
```

```
html-placeholder="State">
```

```
</div>
```

```
<!-- Buttons column - aligned to bottom -->
```

```
<div class="col-md-3" style="text-align: right;">
```

```
<apex:commandButton action="{!search}"
```

```
value="Search"
```

```
styleClass="btn btn-primary"
```

```
style="margin-right: 8px; padding: 10px 20px;">
```

```
<apex:commandButton action="{!clearFilters}"
```

```
value="Clear"
```

```
styleClass="btn btn-default"
```

```
style="margin-right: 8px; padding: 10px 20px;">
```

```
<a href="https://RDG209.github.io/RegistrationForm.html?nocache=1"
```

```
target="_blank"
```

```
class="btn btn-success"
```

```
style="background: #28a745; border: none; padding: 10px 15px;">
```

```
+ Register New
```

```
</a>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- Results Count -->
```

```
<apex:outputText value="Found {!institutions.size} institution(s)"
```

```
styleClass="text-muted"/>
```

```
<!-- Institution List -->
```

```
<apex:repeat value="{!institutions}" var="inst">
```

```
<div class="institution-card">
```

```
<!-- institution image per card, on left -->
```

```
<div style="flex-shrink: 0; width: 120px;">
```

```
<apex:outputPanel rendered="{!(NOT(ISBLANK(inst.Image_URL__c)))}">
```

```

```

```
</apex:outputPanel>
```

```
<apex:outputPanel rendered="{!(ISBLANK(inst.Image_URL__c))}">
```

```
<div style="width: 120px; height: 120px; background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
```

```
border-radius: 8px; display: flex; align-items: center;
```

```
justify-content: center; font-size: 3em;">
```

```
<apex:outputText rendered="{!inst.Category__c == 'Art Gallery/Walk'}"><img alt="Art Gallery/Walk icon" /></apex:outputText>
```

```
<apex:outputText rendered="{!inst.Category__c == 'Cultural Site'}"><img alt="Cultural Site icon" /></apex:outputText>
```

```
<apex:outputText rendered="{!inst.Category__c == 'Garden'}"><img alt="Garden icon" /></apex:outputText>
```

```
<apex:outputText rendered="{!inst.Category__c == 'Library/Books'}"><img alt="Library/Books icon" /></apex:outputText>
```

```
<apex:outputText rendered="{!inst.Category__c == 'Museum'}"><img alt="Museum icon" /></apex:outputText>
```

```
</div>
```

```
</apex:outputPanel>
```

```
</div>
```

```
<div style="flex: 1; min-width: 0;">
```

```
<div class="institution-name">{!inst.Name}</div>
```

```
<div class="category-badge">{!inst.Category__c}</div>
```

```
<div><strong>Location:</strong> {!inst.City__c}, {!inst.State__c} {!inst.Zip__c}</div>
```

```
<div class="contact-info">
```

```
<apex:outputText rendered="{!(NOT(ISBLANK(inst.Phone__c)))}">
```

```
<img alt="Phone icon" /> {!inst.Phone__c} </apex:outputText>
```

```
</apex:outputText>
```

```
<apex:outputText rendered="{!(NOT(ISBLANK(inst.Email__c)))}">
```

```
<img alt="Email icon" /> {!inst.Email__c} </apex:outputText>
```

```
</apex:outputText>
```

```
</div>
```

```
<div class="action-buttons">
```

```
<apex:commandLink action="{!viewDetail}" styleClass="btn btn-info btn-sm">
```

```
<apex:param name="instId" value="{!inst.Id}" />
```

```
View Details
```

```
</apex:commandLink>
```

```
<apex:outputLink value="{!inst.Website__c}" target="_blank"
```

```
styleClass="btn btn-default btn-sm"
```

```
rendered="{!(NOT(ISBLANK(inst.Website__c)))}">
```

```
Website
```

```
</apex:outputLink>
```

```

<apex:outputLink value="{!inst.Email__c}"
    styleClass="btn btn-default btn-sm"
    rendered="{!NOT(ISBLANK(inst.Email__c))}">
    Contact
</apex:outputLink>
<apex:outputLink value="{!inst.Donation__c}" target="_blank"
    styleClass="btn btn-success btn-sm"
    rendered="{!inst.Accepts_Donations__c}">
    Donate
</apex:outputLink>
</div>
<!-- RATINGS -->
<apex:outputPanel rendered="{!inst.Rating_Count__c > 0}">
    <div style="color:#f5a623; font-size:1.1em; margin:4px 0;">
        <apex:outputText value="{!inst.Average_Rating__c}"/> / 5
        ★ ({!inst.Rating_Count__c} ratings)
    </div>
</apex:outputPanel>
<apex:outputPanel rendered="{!inst.Rating_Count__c == 0}">
    <div style="color:#aaa; font-size:0.9em; margin:4px 0;">No ratings yet</div>
</apex:outputPanel>
</div>
</apex:repeat>
<!-- No Results -->
<apex:outputPanel rendered="{!institutions.size == 0}" styleClass="no-results">
    <p>No institutions found</p>
    <p>Try adjusting your search or register a new Institution!</p>
</apex:outputPanel>
</apex:outputPanel>
<!-- DETAIL VIEW -->
<apex:outputPanel rendered="{!showDetail}">
    <apex:commandLink action="{!backToList}"
        styleClass="btn btn-default"
        immediate="true">
        ← Back to Directory
    </apex:commandLink>
    <div class="institution-card" style="margin-top: 20px; display: block;">
        <div class="institution-name">
            {!selectedInstitution.Name}
        </div>
        <div class="category-badge">
            {!selectedInstitution.Category__c}
        </div>
        <!-- detail image -->
        <apex:outputPanel rendered="{!NOT(ISBLANK(selectedInstitution.Image_URL__c))}">
            
        </apex:outputPanel>
        <apex:outputPanel rendered="{!ISBLANK(selectedInstitution.Image_URL__c)}">
            <div style="width:100%; height:120px; background:linear-gradient(135deg,#667eea 0%,#764ba2 100%);
                border-radius:6px; margin-bottom:12px; display:flex; align-items:center;
                justify-content:center; color:white; font-size:2em;">
                <apex:outputText rendered="{!selectedInstitution.Category__c == 'Art Gallery/Walk'}">🎨</apex:outputText>
                <apex:outputText rendered="{!selectedInstitution.Category__c == 'Cultural Site'}">🏛️</apex:outputText>
                <apex:outputText rendered="{!selectedInstitution.Category__c == 'Garden'}">🌿</apex:outputText>
                <apex:outputText rendered="{!selectedInstitution.Category__c == 'Library/Books'}">📖</apex:outputText>
                <apex:outputText rendered="{!selectedInstitution.Category__c == 'Museum'}">🏺</apex:outputText>
            </div>
        </apex:outputPanel>
        <!-- Address -->
        <div style="margin: 15px 0;">
            <b4>Location</b4>
            <p>
                {!selectedInstitution.Street_Address__c}<br/>
                {!selectedInstitution.City__c}, {!selectedInstitution.State__c} {!selectedInstitution.ZIP__c}
            </p>
            </div>
            <!-- MAP CONTAINER (only if has location) -->
            <apex:outputPanel rendered="{!hasLocation}">
                <div style="margin: 15px 0;">
                    <b4>Map</b4>
                    <div id="map" style="height: 300px; width: 100%; border-radius: 5px; border: 1px solid #ddd;"></div>
                </div>
            </apex:outputPanel>
            <!-- Description -->
            <apex:outputPanel rendered="{!NOT(ISBLANK(selectedInstitution.Description__c))}">
                <div style="margin: 15px 0;">
                    <b4>About</b4>
                    <p>{!selectedInstitution.Description__c}</p>
                </div>
            </apex:outputPanel>
            <!-- Contact -->
            <div style="margin: 15px 0;">
                <b4>Contact</b4>
                <p>
                    <apex:outputText rendered="{!NOT(ISBLANK(selectedInstitution.Phone__c))}">
                        Phone: {!selectedInstitution.Phone__c}<br/>
                    <apex:outputText>
                        <apex:outputText rendered="{!NOT(ISBLANK(selectedInstitution.Email__c))}">
                            Email: {!selectedInstitution.Email__c}<br/>
                        <apex:outputText>
                            </p>
                        </div>
                    <!-- Action Buttons -->
                    <div class="action-buttons">
                        <apex:outputLink value="{!selectedInstitution.Website__c}"
                            target="_blank"
                            styleClass="btn btn-primary"
                            rendered="{!NOT(ISBLANK(selectedInstitution.Website__c))}">
                            Visit Website
                        </apex:outputLink>
                        <apex:outputLink value="{!selectedInstitution.Donation__c}"
                            target="_blank"
                            styleClass="btn btn-success"
                            rendered="{!selectedInstitution.Accepts_Donations__c}">
                            Make Donation
                        </apex:outputLink>
                        <apex:outputLink value="mailto:{!selectedInstitution.Email__c}"
                            styleClass="btn btn-default"
                            rendered="{!NOT(ISBLANK(selectedInstitution.Email__c))}">
                            Send Email
                        </apex:outputLink>
                    </div>
                </div>
            </div>
            <!-- Leadership Staff -->
            <apex:outputPanel rendered="{!staffMembers.size > 0}">
                <b3 style="margin-top: 30px;">Leadership</b3>
                <apex:repeat value="{!staffMembers}" var="staff">
                    <div class="staff-card">

```

```

<div style="flex-shrink:0;">
<apex:outputPanel rendered="{!(NOT(ISBLANK(staff.Photo__c)))}">
  
</apex:outputPanel>
<apex:outputPanel rendered="{!(ISBLANK(staff.Photo__c))}">
  <div style="width:70px; height:70px; border-radius:50%;
    background:linear-gradient(135deg,#667eea 0%,#764ba2 100%);
    display:flex; align-items:center; justify-content:center;
    color:white; font-size:1.5em; border:3px solid #667eea;">👤</div>
</apex:outputPanel>
</div>

<div style="flex:1;">
<div style="font-weight:bold; font-size:1.1em;">{!staff.Name}</div>
<div style="color:#666; font-style:italic;">{!staff.Role__c}</div>
<apex:outputPanel rendered="{!(NOT(ISBLANK(staff.Bio__c)))}">
  <p style="margin-top:8px;">{!staff.Bio__c}</p>
</apex:outputPanel>
<apex:outputLink value="mailto:{!staff.Email__c}" rendered="{!(NOT(ISBLANK(staff.Email__c)))}">
  Contact {!staff.Name}
</apex:outputLink>
</div>

</div>

<apex:repeat>
<apex:outputPanel>

<apex:outputPanel rendered="{!staffMembers.size == 0}">
  <p style="color: #666; margin-top: 20px;">No leadership information available.</p>
</apex:outputPanel>

<!-- RATINGS -->
<div style="margin-top:25px; padding:20px; background:#f8f9fa; border-radius:8px; border:1px solid #ddd;">
  <h4 style="margin-top:0;">Rate This Institution</h4>

  <!-- Display current average -->
  <apex:outputPanel rendered="{!selectedInstitution.Rating_Count__c > 0}">
    <div style="font-size:1.3em; color:#5a623; margin-bottom:15px;">
      ★ <apex:outputText value="{!selectedInstitution.Average_Rating__c}" /> / 5
    <span style="color:#666; font-size:0.7em;">
      ({!selectedInstitution.Rating_Count__c} ratings)
    </span>
  </div>
</apex:outputPanel>
<apex:outputPanel rendered="{!selectedInstitution.Rating_Count__c == 0}">
  <div style="color:#aaa; margin-bottom:15px;">No ratings yet — be the first!</div>
</apex:outputPanel>

<!-- Messages -->
<apex:messages styleClass="alert alert-info" style="margin-bottom:10px;">

<!-- Rating form -->
<div style="display:flex; gap:12px; align-items:flex-end; flex-wrap:wrap;">
  <div>
    <label style="display:block; margin-bottom:4px;">Your Name (optional)</label>
    <apex:inputText value="{!reviewerName}"
      styleClass="form-control"
      style="width:200px;"
      html-placeholder="Anonymous" />
  </div>
  <div>
    <label style="display:block; margin-bottom:4px;">Rating</label>
    <apex:selectList value="{!selectedRating}" size="1" styleClass="form-control"
      style="width:150px;">
    <apex:selectOptions value="{!starOptions}" />
  </div>
  <div>
    <apex:selectList>
  </div>
  <div>

```

```

<apex:commandButton action="{!submitRating}"
  value="Submit Rating"
  styleClass="btn btn-primary"
  style="padding:10px 20px;" />
</div>
</div>
</div>
</apex:outputPanel>

<!-- MAP JAVASCRIPT - MOVED OUTSIDE FORM, ONLY LOADS WHEN DETAIL SHOWN -->
<apex:outputPanel rendered="{!showDetail && hasLocation}">
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
  <script>
    // Wait for DOM to be ready
    document.addEventListener('DOMContentLoaded', function() {
      var lat = {!selectedLat};
      var lng = {!selectedLng};

      // Check if map container exists
      var mapContainer = document.getElementById('map');
      if (!mapContainer) {
        console.error('Map container not found');
        return;
      }

      // Initialize map
      var map = L.map('map').setView([lat, lng], 15);

      // Add OpenStreetMap tiles
      L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
        attribution: '&copy; OpenStreetMap contributors',
        maxZoom: 19
      }).addTo(map);

      // Add marker
      L.marker([lat, lng]).addTo(map)
        .bindPopup('{!$ENCODE(selectedInstitution.Name)}')
        .openPopup();
    });
  </script>
</apex:outputPanel>

</apex:form>
</div>

</apex:page>

```

InstitutionDirectoryController (main driver)

```

public class InstitutionDirectoryController {

  // Search and filter inputs
  public String searchName { get; set; }
  public String filterCategory { get; set; }
  public String filterCity { get; set; }
  public String filterState { get; set; }

  // Rating submission fields
  public String reviewerName { get; set; }
  public Integer selectedRating { get; set; }

  // Results
  public List<Institution__c> institutions { get; set; }
  public Institution__c selectedInstitution { get; set; }
  public List<Staff_Member__c> staffMembers { get; set; }
  public Boolean showDetail { get; set; }

  // Picklist options for filters
  public List<SelectOption> getCategoryOptions() {
    List<SelectOption> options = new List<SelectOption>();
    options.add(new SelectOption('All Categories'));

    // Get picklist values dynamically
    Schema.DescribeFieldResult fieldResult = Institution__c.getDescribe();
    List<Schema.PicklistEntry> ple = fieldResult.getPicklistValues();
    for(Schema.PicklistEntry f: ple) {
      options.add(new SelectOption(f.getLabel(), f.getValue()));
    }
  }
}

```

```

        return options;
    }

    // Constructor - load initial data
    public InstitutionDirectoryController() {
        showDetail = false;
        loadAllInstitutions();
    }

    // Load all institutions (up to 50)
    public void loadAllInstitutions() {
        institutions = (SELECT Id, Name, Category__c, City__c, State__c,
            Street_Address__c, ZIP__c, Phone__c, Email__c,
            Website__c, Donation__c, Accepts_Donations__c,
            Description__c, Location__c, Image_URL__c, Average_Rating__c, Rating_Count__c
            FROM Institution__c
            WHERE Status__c = 'Active'
            ORDER BY Name
            LIMIT 50);
    }

    // Search and filter action
    public PageReference search() {
        String query = 'SELECT Id, Name, Category__c, City__c, State__c, ' +
            'Street_Address__c, ZIP__c, Phone__c, Email__c, ' +
            'Website__c, Donation__c, Accepts_Donations__c, ' +
            'Description__c, Location__c, Image_URL__c, Average_Rating__c, Rating_Count__c ' +
            'FROM Institution__c ' +
            'WHERE Status__c = \'Active\'';

        // Add name search
        if (String.isNotBlank(searchName)) {
            String searchTerm = '%' + String.escapeSingleQuotes(searchName) + '%';
            query += 'AND Name LIKE \'' + searchTerm + '\'';
        }

        // Add category filter
        if (String.isNotBlank(filterCategory)) {
            query += 'AND Category__c = \'' + String.escapeSingleQuotes(filterCategory) + '\'';
        }

        // Add city filter
        if (String.isNotBlank(filterCity)) {
            String cityTerm = '%' + String.escapeSingleQuotes(filterCity) + '%';
            query += 'AND City__c LIKE \'' + cityTerm + '\'';
        }

        // Add state filter
        if (String.isNotBlank(filterState)) {
            query += 'AND State__c = \'' + String.escapeSingleQuotes(filterState) + '\'';
        }

        query += 'ORDER BY Name LIMIT 50';

        try {
            institutions = Database.query(query);
        } catch (Exception e) {
            ApexPages.addMessage(new ApexPages.Message(ApexPages.Severity.ERROR,
                'Error searching: ' + e.getMessage()));
            institutions = new List<Institution__c>();
        }

        return null; // Stay on page
    }

    // View institution detail
    public PageReference viewDetail() {
        String instId = ApexPages.currentPage().getParameters().get('instId');

        if (String.isNotBlank(instId)) {
            try {
                selectedInstitution = (SELECT Id, Name, Category__c, City__c, State__c,
                    Street_Address__c, ZIP__c, Phone__c, Email__c,
                    Website__c, Donation__c, Accepts_Donations__c,
                    Description__c, Location__c, Image_URL__c, Average_Rating__c, Rating_Count__c
                    FROM Institution__c
                    WHERE Id = :instId
                    LIMIT 1);

                // Load associated staff
                staffMembers = (SELECT Id, Name, Role__c, Email__c, Bio__c,
                    Photo__c, Is_Leadership__c
                    FROM Staff_Member__c
                    WHERE Institution__c = :instId
                    AND Is_Leadership__c = true
                    ORDER BY Name);

                showDetail = true;
            } catch (Exception e) {
                ApexPages.addMessage(new ApexPages.Message(ApexPages.Severity.ERROR,
                    'Institution not found.));
            }
        }

        return null;
    }

    // Back to list
    public PageReference backToList() {
        showDetail = false;
        selectedInstitution = null;
        staffMembers = null;
        return null;
    }

    // Get full address as single string
    public String getFullAddress(Institution__c inst) {
        if (inst == null) return '';

        List<String> parts = new List<String>();
        if (String.isNotBlank(inst.Street_Address__c)) parts.add(inst.Street_Address__c);
        if (String.isNotBlank(inst.City__c)) parts.add(inst.City__c);
        if (String.isNotBlank(inst.State__c)) parts.add(inst.State__c);
        if (String.isNotBlank(inst.ZIP__c)) parts.add(inst.ZIP__c);

        return String.join(parts, ',');
    }

    // Clear all filters
    public PageReference clearFilters() {
        searchName = '';
        filterCategory = '';
        filterCity = '';
        filterState = '';
        loadAllInstitutions();
        return null;
    }

    // Map methods for coords
    public Double getSelectedLat() {
        if (selectedInstitution == null || selectedInstitution.Location__c == null) {
            return 34.0522; // Default: Los Angeles
        }
        return selectedInstitution.Location__c.getLatitude();
    }

    public Double getSelectedLng() {
        if (selectedInstitution == null || selectedInstitution.Location__c == null) {
            return -118.2437; // Default: Los Angeles
        }
        return selectedInstitution.Location__c.getLongitude();
    }

    public Boolean getHasLocation() {
        return selectedInstitution != null && selectedInstitution.Location__c != null;
    }

    // Submit a rating
    public PageReference submitRating() {
        if (selectedInstitution == null) return null;

        if (selectedRating == null || selectedRating < 1 || selectedRating > 5) {
            ApexPages.addMessage(new ApexPages.Message(
                ApexPages.Severity.ERROR, 'Please select a rating between 1 and 5.));
            return null;
        }

        try {
            Institution_Rating__c rating = new Institution_Rating__c(
                Institution__c = selectedInstitution.Id,
                Rating__c = selectedRating,
                Reviewer_Name__c = String.isNotBlank(reviewerName) ? reviewerName : 'Anonymous'
            );
            insert rating;

            // Re-query to get updated roll-up values
            selectedInstitution = (SELECT Id, Name, Category__c, City__c, State__c,
                Street_Address__c, ZIP__c, Phone__c, Email__c,
                Website__c, Donation__c, Accepts_Donations__c,
                Description__c, Location__c, Image_URL__c,
                Average_Rating__c, Rating_Count__c
                FROM Institution__c
                WHERE Id = :selectedInstitution.Id LIMIT 1);

            reviewerName = '';
            selectedRating = null;

            ApexPages.addMessage(new ApexPages.Message(
                ApexPages.Severity.CONFIRM, 'Thank you for your rating!));
            loadAllInstitutions();
        } catch (Exception e) {
            ApexPages.addMessage(new ApexPages.Message(
                ApexPages.Severity.ERROR, 'Error saving rating: ' + e.getMessage()));
        }

        return null;
    }

    // Star options for the select list
    public List<SelectOption> getStarOptions() {
        List<SelectOption> options = new List<SelectOption>();
        options.add(new SelectOption('1', 'Select 1'));
        options.add(new SelectOption('2', '★ 2'));
        options.add(new SelectOption('3', '★★ 3'));
        options.add(new SelectOption('4', '★★★ 4'));
        options.add(new SelectOption('5', '★★★★★ 5'));
        return options;
    }
}

```

InstitutionGeocodeHelper (helper class)

```

public class InstitutionGeocodeHelper {

    @future(callout=true)
    public static void geocodeAsync(Id institutionId) {

        // Re-query the record - include latitude/longitude fields
        Institution__c inst = {
            SELECT Id, Street_Address__c, City__c, State__c, ZIP__c,
                Location__Latitude__s, Location__Longitude__s
            FROM Institution__c
            WHERE Id = :institutionId
            LIMIT 1
        };

        // Build address string
        List<String> addressParts = new List<String>();
        if (String.isNotBlank(inst.Street_Address__c)) addressParts.add(inst.Street_Address__c);
        if (String.isNotBlank(inst.City__c)) addressParts.add(inst.City__c);
        if (String.isNotBlank(inst.State__c)) addressParts.add(inst.State__c);
        if (String.isNotBlank(inst.ZIP__c)) addressParts.add(inst.ZIP__c);

        String fullAddress = String.join(addressParts, ',');
        String encodedAddress = EncodingUtil.urlEncode(fullAddress, 'UTF-8');

        // Call OpenStreetMap Nominatim
        Http http = new Http();
        HttpRequest request = new HttpRequest();
        request.setEndpoint('https://nominatim.openstreetmap.org/search?format=json&q=' + encodedAddress);
        request.setMethod('GET');
        request.setRequestHeader('User-Agent', 'HumanitiesHub/1.0 (your@email.com)');
        request.setTimeout(10000);

        try {
            HttpResponse response = http.send(request);

            if (response.getStatusCode() == 200) {
                List<Object> results = (List<Object>) JSON.deserializeUntyped(response.getBody());
            }
        }
    }
}

```



```

if (results.isEmpty()) {
    Map<String, Object> firstResult = (Map<String, Object>) results[0];
    Double lat = Double.valueOf(firstResult.get('lat'));
    Double lng = Double.valueOf(firstResult.get('lon'));

    // CORRECTED: Set latitude and longitude fields directly
    inst.Location__Latitude__s = lat;
    inst.Location__Longitude__s = lng;

    update inst;

    System.debug('Geocoded: ' + fullAddress + ' → ' + lat + ', ' + lng);
} else {
    System.debug('No geocoding results for: ' + fullAddress);
}
} else {
    System.debug('Geocoding API error: ' + response.getStatusCode());
}
}

} catch (Exception e) {
    System.debug('Geocoding exception: ' + e.getMessage());
}
}
}

```

InstitutionGeocode (trigger)

```

trigger InstitutionGeocode on Institution__c (after insert, after update) {

    List<Id> toGeocode = new List<Id>();

    for (Institution__c inst : Trigger.new) {

        Boolean hasAddress = String.isNotBlank(inst.Street_Address__c)
            && String.isNotBlank(inst.City__c)
            && String.isNotBlank(inst.State__c);

        Boolean locationEmpty = inst.Location__c == null;

        Boolean addressChanged = false;
        if (Trigger.isUpdate) {
            Institution__c old = Trigger.oldMap.get(inst.Id);
            addressChanged = (inst.Street_Address__c != old.Street_Address__c)
                || (inst.City__c != old.City__c)
                || (inst.State__c != old.State__c)
                || (inst.ZIP__c != old.ZIP__c);
        }

        if (hasAddress && (Trigger.isInsert || locationEmpty || addressChanged)) {
            toGeocode.add(inst.Id);
        }
    }

    // Call future method once per record outside the loop
    for (Id instId : toGeocode) {
        InstitutionGeocodeHelper.geocodeAsync(instId);
    }
}

```

LeadAutomation (trigger)

```

trigger LeadAutomation on Lead (before insert, after insert) {

    if (Trigger.isAfter && Trigger.isInsert) {
        List<Institution__c> newInstitutions = new List<Institution__c>();
        List<Messaging.SingleEmailMessage> emails = new List<Messaging.SingleEmailMessage>();

        for (Lead l : Trigger.new) {
            if (l.LeadSource == 'Website' || l.LeadSource == 'Web') {
                Institution__c inst = new Institution__c(
                    Name = l.Company,
                    Street_Address__c = l.Street,
                    City__c = l.City,
                    State__c = l.State,
                    ZIP__c = l.PostalCode,
                    Phone__c = l.Phone,
                    Email__c = l.Email,
                    Website__c = l.Website,
                    Description__c = l.Description,
                    Category__c = l.Category__c,
                    Donation__c = l.Donation__c,
                    Status__c = 'Pending'
                );
                newInstitutions.add(inst);

                // Confirmation email to submitter
                if (l.Email != null) {
                    Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
                    mail.setToAddresses(new String[] { l.Email });
                    mail.setSubject('Humanities Hub Registration Received');
                    mail.setPlainTextBody(
                        'Dear ' + l.LastName + ',\n\n' +
                        'Thank you for registering ' + l.Company + ' with Humanities Hub.\n\n' +
                        'Your submission is under review. We will notify you when your organization appears in our directory.\n\n' +
                        'Humanities Hub Team'
                    );
                    emails.add(mail);
                }
            }
        }

        if (newInstitutions.isEmpty()) {
            insert newInstitutions;
        }

        if (emails.isEmpty()) {
            try {
                Messaging.sendEmail(emails);
            } catch (Exception e) {
                System.debug('Email send failed: ' + e.getMessage());
            }
        }
    }
}

```